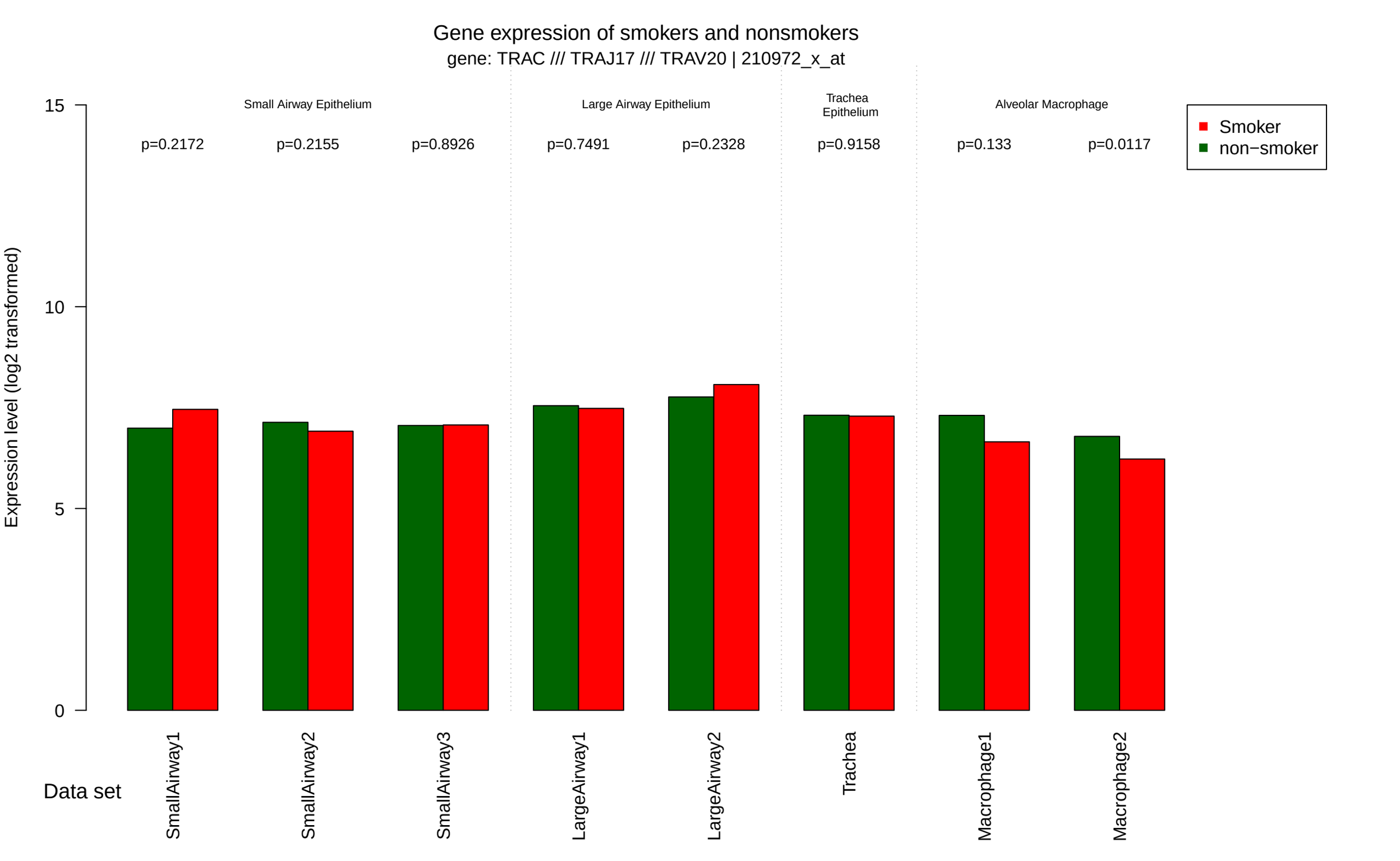
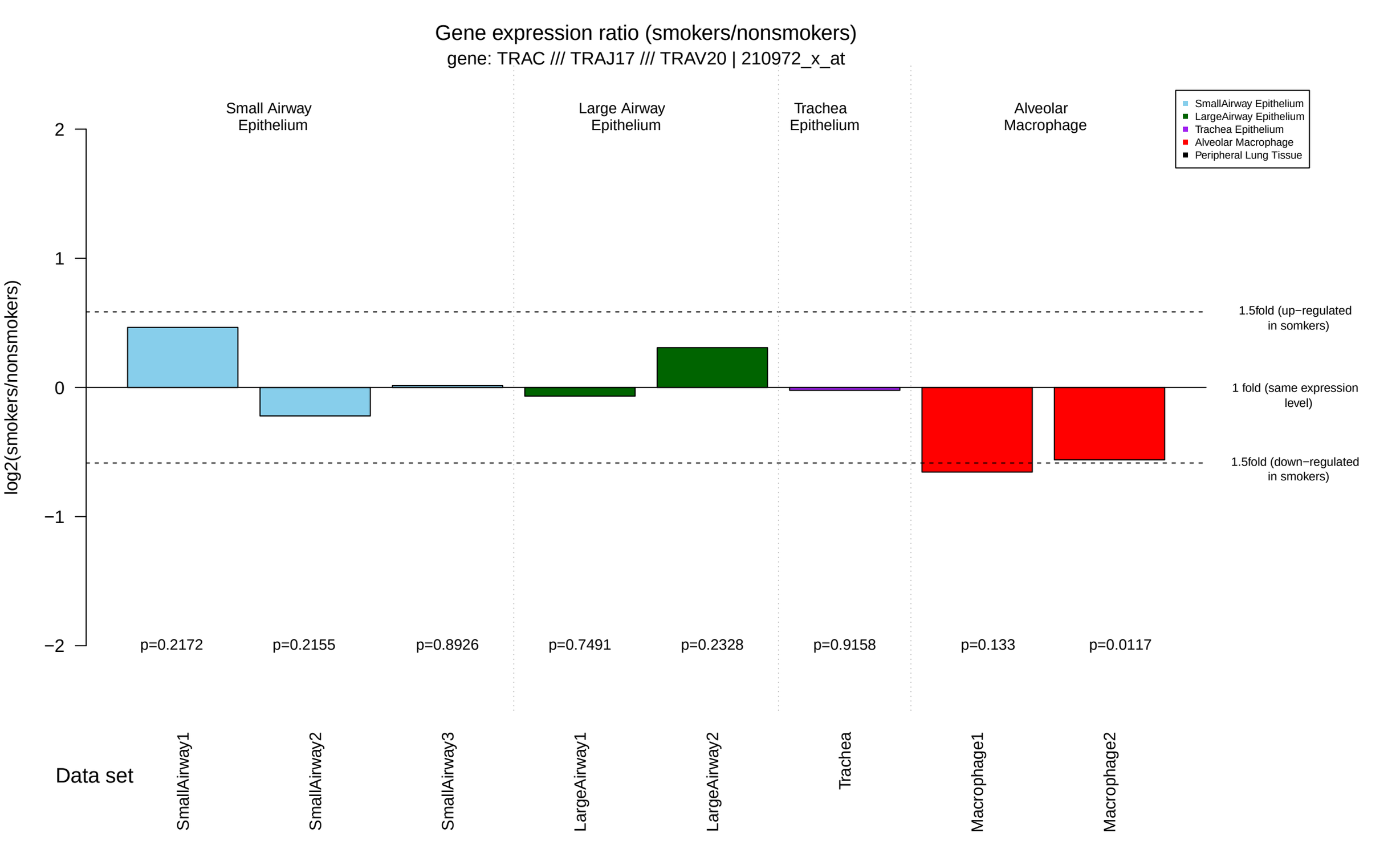
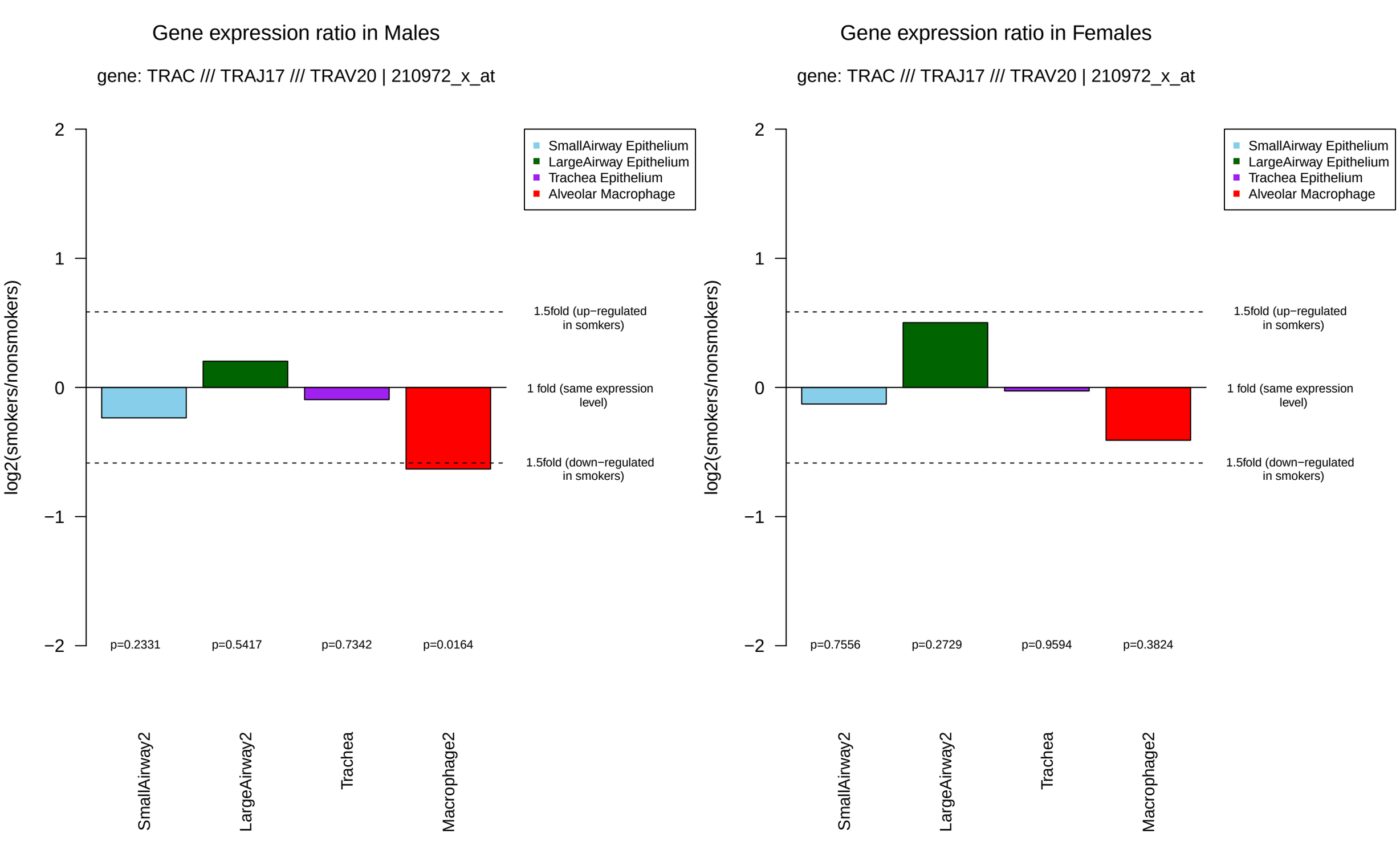
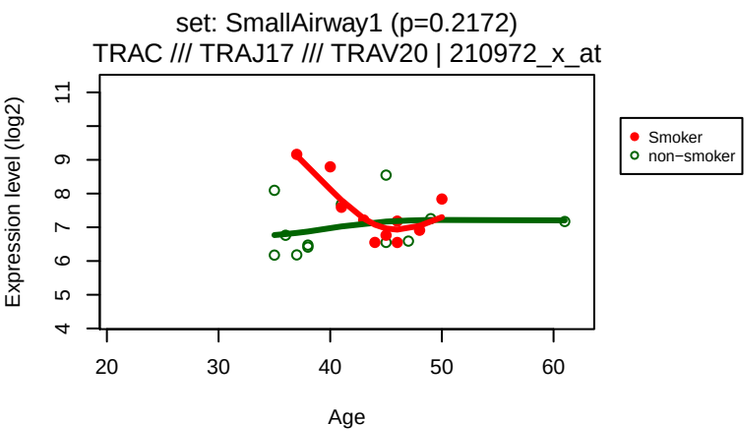
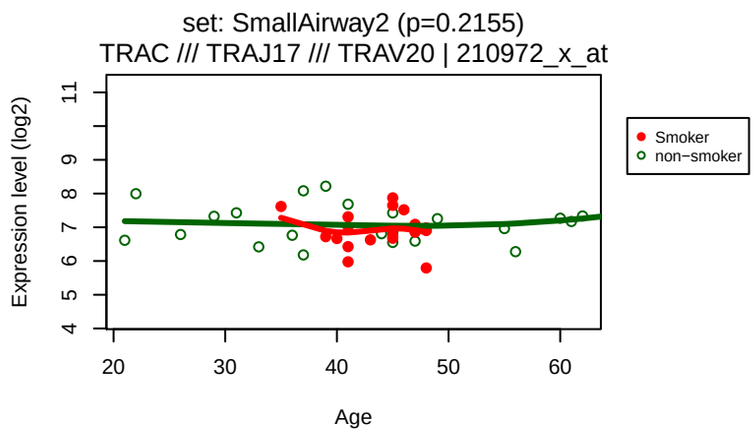
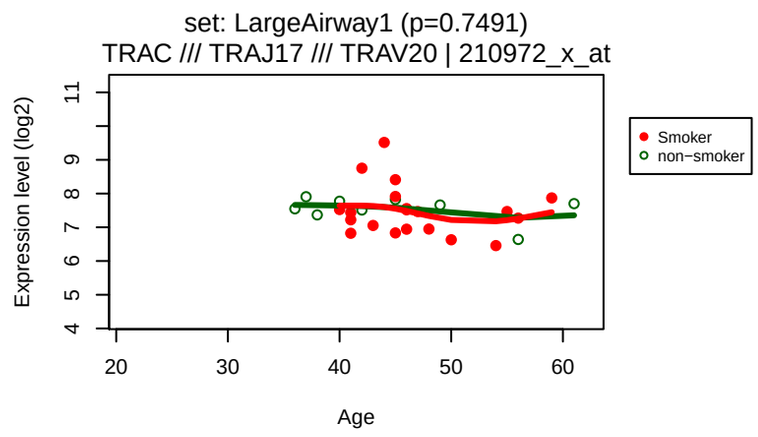

Gene expression of smokers and nonsmokers
gene: TRAC /// TRAJ17 /// TRAV20 | 210972_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2172
p=0.2155
p=0.8926
p=0.7491
p=0.2328
p=0.9158
p=0.133
p=0.0117
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

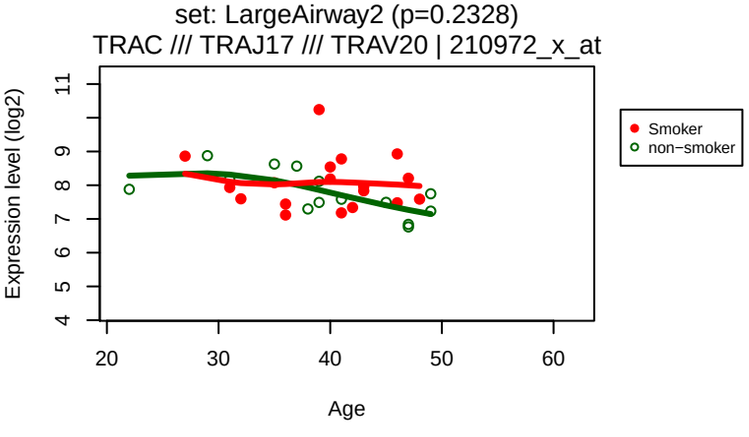

Gene expression ratio (smokers/nonsmokers)
gene: TRAC /// TRAJ17 /// TRAV20 | 210972_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2172
p=0.2155
p=0.8926
p=0.7491
p=0.2328
p=0.9158
p=0.133
p=0.0117
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

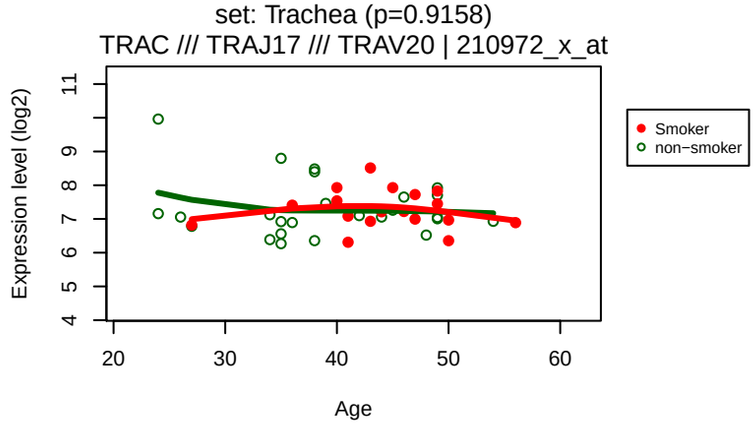

Gene expression ratio in Males
gene: TRAC /// TRAJ17 /// TRAV20 | 210972_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2331
p=0.5417
p=0.7342
p=0.0164
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRAC /// TRAJ17 /// TRAV20 | 210972_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7556
p=0.2729
p=0.9594
p=0.3824
SmallAirway2
LargeAirway2
Trachea
Macrophage2

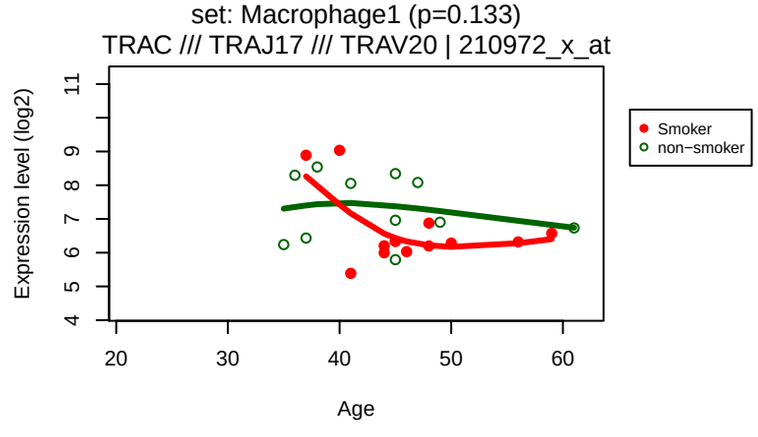

set: SmallAirway1 (p=0.2172)
TRAC /// TRAJ17 /// TRAV20 | 210972_x_at
Expression level (log2)
Age
Smoker
non−smoker

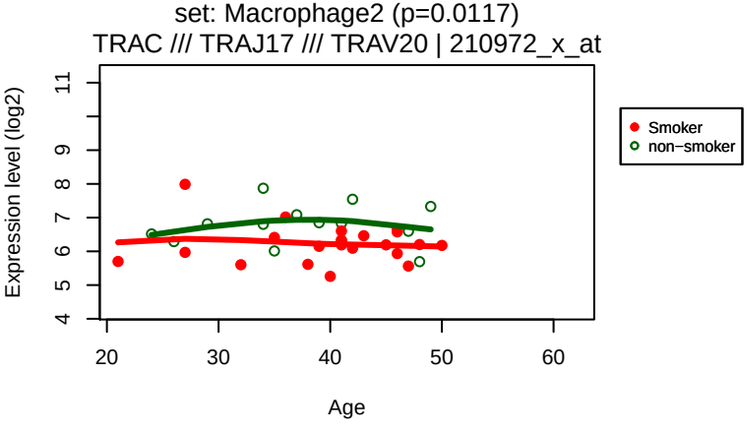

set: SmallAirway2 (p=0.2155)
TRAC /// TRAJ17 /// TRAV20 | 210972_x_at
Expression level (log2)
Age
Smoker
non−smoker

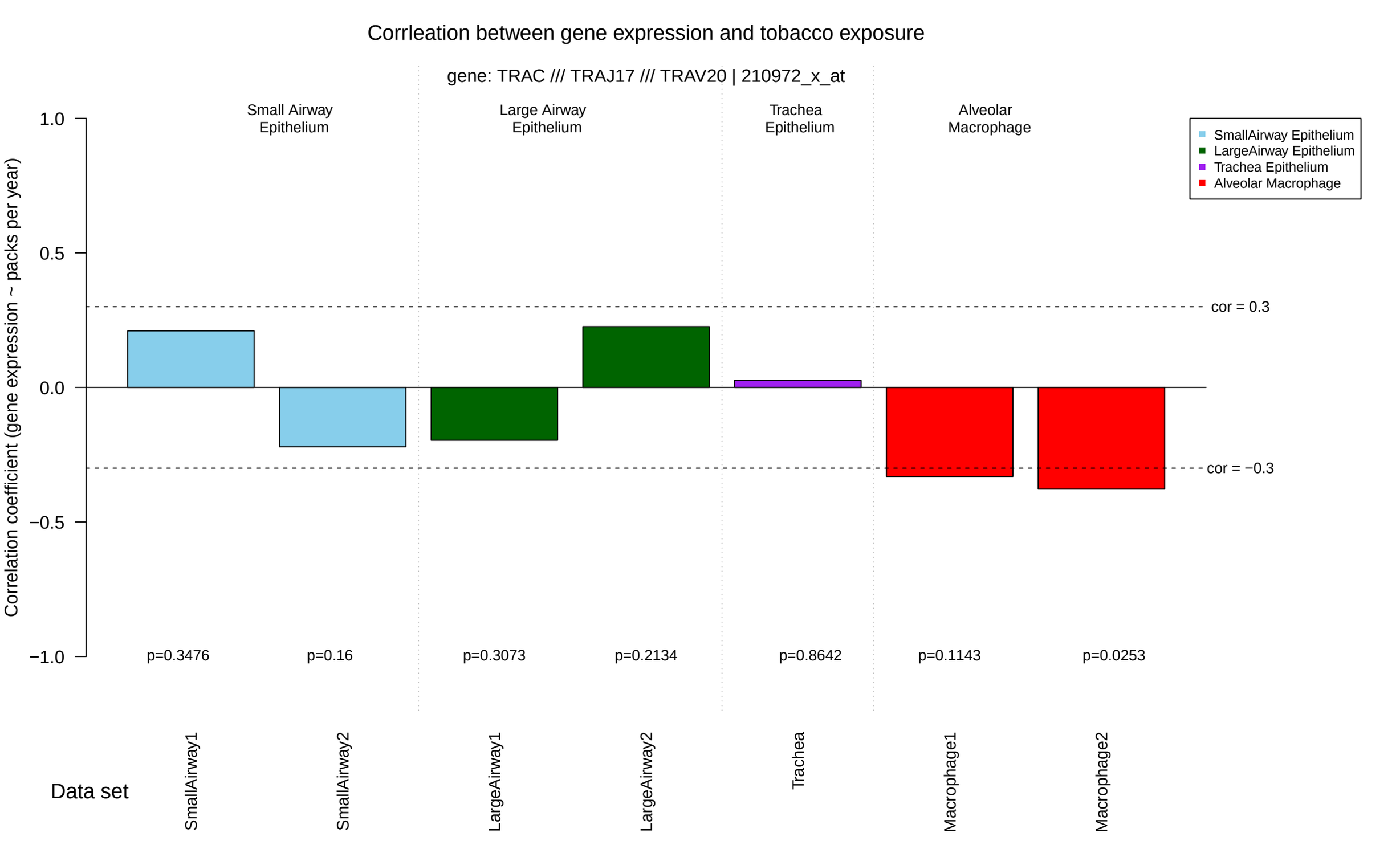

set: LargeAirway1 (p=0.7491)
TRAC /// TRAJ17 /// TRAV20 | 210972_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2328)
TRAC /// TRAJ17 /// TRAV20 | 210972_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9158)
TRAC /// TRAJ17 /// TRAV20 | 210972_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.133)
TRAC /// TRAJ17 /// TRAV20 | 210972_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0117)
TRAC /// TRAJ17 /// TRAV20 | 210972_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRAC /// TRAJ17 /// TRAV20 | 210972_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3476
p=0.16
p=0.3073
p=0.2134
p=0.8642
p=0.1143
p=0.0253
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2